

Shiyao Sang (Samuel)

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IEEE RAS Technical Committee on Cognitive Robotics — Member

Research Focus: Agent systems for robotics · Latent world models · Deep reinforcement learning. My research explores how robots organize information to support reasoning, decision-making, and action, with recent work on task-driven sparse latent planning for autonomous driving and distributed robotic middleware (first-author paper published at IROS 2025).

Education

University of Wollongong, NSW, Australia *M.Sc., Computer Science*
Faculty of Engineering and Information Sciences 2020–2022
Joint Dual-Degree Program with Central China Normal University
Central China Normal University, Wuhan, China *M.Eng., Computer Technology*
Faculty of Artificial Intelligence in Education 2020–2022
Joint Dual-Degree Program with University of Wollongong
Huaiyin Institute of Technology (now Huai'an University), Jiangsu, China *B.Eng., Computer Science and Technology*
Faculty of Computer and Software Engineering 2014–2018

Publications & Research Outputs

- **Shiyao Sang**, Shenghai Yuan, Yuzhi Lai, et al. “Task-Driven Sparse Latent Planning without Future World-State Supervision for Autonomous Driving.” *Manuscript in preparation*.
- **Shiyao Sang**, Yinggang Ling. “Service Discovery-Based Hybrid Network Middleware for Efficient Communication in Distributed Robotic Systems.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. Video: <https://youtu.be/uMvyOykTlka>
- **Shiyao Sang**. “Design and Implementation of a State Transition Model for Educational Robot Tutoring Math Homework.” *Master’s Thesis*, University of Wollongong & Central China Normal University, 2022.
- **Shiyao Sang**. “A Multi-Layer Architecture for Intelligent Chatbots in IoT Robots: Design and Implementation.” *Bachelor’s Thesis*, Huaiyin Institute of Technology, 2018.

Professional Experience

Mogo.ai Information Technology Co., Ltd. Jul 2026 – Present
Senior Algorithm Engineer, Joint Spatiotemporal Planning Group

- Developed multi-agent game-theoretic planning and interaction rollout methods for responsive merging and crossing negotiation in autonomous small-bus driving.

BYD Company Ltd. – New Technology Research Institute Jun 2025 – Dec 2025
Senior Algorithm Engineer, Urban Joint Game Group

- Designed and implemented joint game-theoretic, interaction-aware planning strategies for complex urban scenarios in BYD’s urban intelligent driving system.
- Conducted preliminary research on diffusion-generation-based game-theoretic planning, including baseline reproduction, technical scheme design, and prototype preparation.

Chery Automobile Co., Ltd. – Intelligentization Center Mar 2023 – Mar 2025
Algorithm Engineer, Foresight Algorithm Group May 2024 – Mar 2025

- Worked on lateral decision-making lattice sampling and planning software architecture for an advanced autonomous driving planning framework.
- Proposed a reinforcement-learning-based pruning search method for mapless intersection path generation, recognized as a Chery Group Class-A patent.

Software Engineer, Operating System Department

Mar 2023 – May 2024

- Designed and implemented communication middleware deployed on Robotaxi and Robotruck platforms; the related work was published as a first-author paper at IROS 2025.
- Designed a multi-Orin distributed computing architecture for Robotaxi, balancing algorithm deployment across computing resources and communication loads.

Coowa Technology Co., Ltd.

Jul 2022 – Jan 2023

Software Engineer, Remote Driving Group

- Developed network security and cockpit audio modules for an L4 autonomous vehicle teleoperation system; contributed an encryption-related fix MR to ZLMediaKit.

Patents

- **Shiyao Sang.** “An Adaptive Trajectory Generation Method for Intersections Without High-Precision Maps Based on Multi-Deciders and Evaluators.” *Invention Patent*, 2024.
- **Shiyao Sang.** “Digital Clock-based Automatic Latency Testing Method and System for Remote Driving Streaming Media.” *Invention Patent*, 2022.

Technical Skills

- **AI & Learning:** Latent World Models, Deep Learning, Reinforcement Learning, Markov Decision Processes (MDP), Monte Carlo Tree Search (MCTS), and VLM-based Planning.
- **Robotic Middleware & Systems:** Expertise in distributed computing architectures (ROS, ROS2, CyberRT) and SOME/IP software architecture design.
- **Human-Robot Interaction (HRI):** Dialogue system design (AIML), prompt engineering, and LLM fine-tuning for interactive robotics.
- **Programming:** C++, C#, Python; PyTorch and TensorFlow.

Honors & Awards

- 2017 “Zhongxing Cup” Pan-Pearl River Delta University Computer Works Competition — Silver (Undergraduate Group)
- 2017 China Engineering Robot Competition — IoT Robot Innovation — Second Prize
- 2018 Microsoft Imagine Cup — Suzhou Regional Final — Third Prize
- Jiangsu Province 5th Science Students Humanities & Social Science Knowledge Contest — Excellence Award

Referees

Dr. Shenghai Yuan – Research Collaborator – Senior Research Fellow, Nanyang Technological University – Shyuan@ntu.edu.sg

Dr. Zhilong Liu – Former Technical Supervisor – Senior Algorithm Engineer, Woven by Toyota – lzl200102109@gmail.com